

Y2 Quantum Mechanics

Part 1 – Wave Mechanics

Observables; operators; wave theory of matter; Schrödinger equation; interpreting the wave-function and expectation values.

Part 2 – Quantum Mechanics in 1D

Solutions of the time-independent Schrödinger equation: potential barriers; quantum tunnelling; square well potentials; simple harmonic oscillator potential.

Part 3 – Principles of Quantum Mechanics

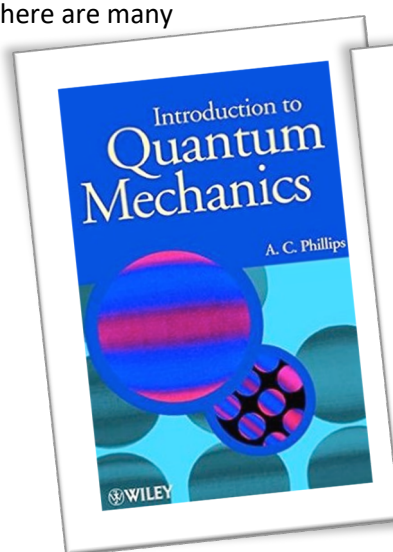
Time dependence; superposition and measurement; simultaneous eigenfunctions; commutation and uncertainty.

Part 4 – Quantum Mechanics in 3D

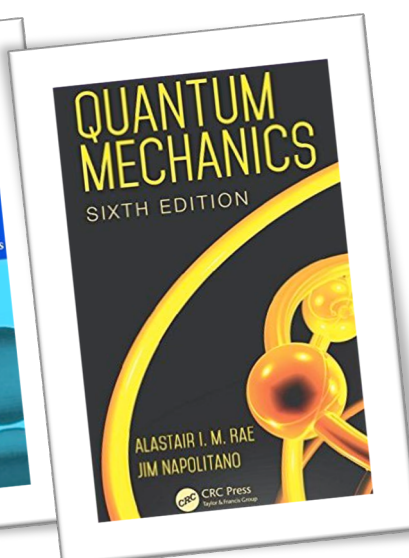
Degeneracy; angular momentum; central potentials; single-electron atoms (hydrogen); single-electron atom wave-functions; Stern-Gerlach experiment – intrinsic spin; total angular momentum; atomic structure; Pauli exclusion principle; Ladder operators.

Textbooks

There are many



Good level for this course



More advanced

See also:

Quantum Physics of Atoms, Molecules, Solids, Nuclei and Particles, Eisberg and Resnick

Quantum Physics, S. Gasiorowicz

Quantum Mechanics is a philosophically challenging subject:

However, most were still unhappy with what they had discovered ...

'Anyone who is not shocked by quantum theory has not understood it.'

Niels Bohr

'I think I can safely say that nobody understands quantum mechanics.'

Richard Feynman

... a some were never reconciled to it

'I am, in fact, firmly convinced that the essentially statistical character of contemporary quantum theory is solely to be ascribed to the fact that this (theory) operates with an incomplete description of physical reality.' **Albert Einstein**

Y1 Quantum Mechanics introduced:

- Blackbody radiation and atomic spectra
- Photoelectric effect
- Compton scattering
- Electron Diffraction
- Photons and electrons can behave both as waves and particles, e.g. from the photoelectric effect and diffraction
- Waves can interfere and superpose
- Energy is proportional to frequency $E = h\nu$
- Momentum is inversely proportional to wavelength $p = h/\lambda$ [de Broglie PhD 1924, Nobel Prize 1929]

Y2 QM will develop the formalism of QM

1. Principles of Quantum Mechanics

The last section focussed on solutions of the TISE.

$$\hat{T}\psi_n + \hat{V}\psi_n = E_n\psi_n$$

Or

$$\hat{H}\psi_n = E_n\psi_n$$

Where $\hat{H}$ is the total energy operator, called the Hamiltonian.

ψ_n is an eigenfunction of the operator $\hat{H}$, and E_n is an associated energy eigenvalue.

The full time-dependent wavefunction is

$$\Psi_n(x, t) = \psi_n(x)e^{-iE_nt/\hbar}$$

$$|\Psi_n|^2 = \psi_n(x)^* e^{iE_nt/\hbar} \psi_n(x) e^{-iE_nt/\hbar} = |\psi_n|^2$$

$|\psi_n|^2$ has no time dependence and hence $\Psi_n(x, t)$ describes a *stationary state*, i.e. independent of time and solutions of SE and TISE give same outcome.

Now consider a mixed state

$$\Phi(x, t) = \Psi_n(x, t) + \Psi_m(x, t)$$

$$|\Phi|^2 = (\Psi_n(x, t)^* + \Psi_m(x, t)^*)(\Psi_n(x, t) + \Psi_m(x, t))$$

$$|\Phi|^2 = \Psi_n^* \Psi_n + \Psi_m^* \Psi_m + \underbrace{\Psi_m^* \Psi_n + \Psi_n^* \Psi_m}_{\text{interference terms}}$$

$$\Psi_m^* \Psi_n = \psi_m^* \psi_n e^{i(E_m - E_n)t/\hbar} = z$$

$$\Psi_n^* \Psi_m = \psi_n^* \psi_m e^{i(E_n - E_m)t/\hbar} = z^*$$

$$\Delta E = (E_m - E_n)$$

$$z = \psi_m^* \psi_n \left[\cos\left(\frac{\Delta E}{\hbar} t\right) + i \sin\left(\frac{\Delta E}{\hbar} t\right) \right]$$

$$\text{Noting that } z + z^* = 2\text{Re}\{z\}$$

$$|\Phi|^2 = |\psi_n|^2 + |\psi_m|^2 + 2\psi_m^* \psi_n \cos\left(\frac{\Delta E t}{\hbar}\right)$$

Where $\psi_m^* \psi_n = \psi_n^* \psi_m$ (and we will assume real)

Now the probability density has a time dependence and oscillates with an angular frequency $\omega = \Delta E/\hbar = (E_m - E_n)/\hbar$



Above can see the result of the two lowest energy SHO levels interfering which results in a time dependence of the probability density $|\Phi|^2$. $\Phi(x, t)$ is an example of a superposition of states.

If $\Psi_n(x, t)$ and $\Psi_m(x, t)$ are solutions of the SE, then so is $\Phi(x, t) = c_n \Psi_n(x, t) + c_m \Psi_m(x, t)$ since the SE is a linear ODE.

But $\phi(x) = \psi_n(x) + \psi_m(x)$ is not a solution to the TISE, as total energy, E , of $\psi_n(x) \neq \psi_m(x)$.

Demonstration:

$$\hat{H}\phi = \hat{H}\psi_n + \hat{H}\psi_m = E_m\psi_m + E_n\psi_n \neq E(\psi_m + \psi_n) \quad [\text{unless } E_n = E_m]$$

In general $\phi(x) = \psi_n(x) + \psi_m(x)$ is not an eigenfunction of $\hat{H}$ – it has no definite value of energy.

What does this mean!

Need to understand superposition and measurement.....

Superposition and Measurement

Let $\Psi_i(x, t)$ ($i = 1, 2, \dots, n$) form a complete set of solutions to the Schrödinger Equation.

Then a general solution can be expressed as a linear combination of these solutions

$$\Phi(x, t) = \sum_{i=1}^n c_i \Psi_i(x, t)$$

Here c_i are a set of coefficients which could be real or complex.

Postulate 3 Any wavefunction can be written as

$$\Phi(x, t) = \sum_{i=1}^n c_i \Psi_i(x, t)$$

In general $\Phi(x, t)$ is not an eigenfunction of $\hat{H}$ (or other QM operators) and so does not have a definite value of E , so what is the result of the measurement of $\Phi(x, t)$?

If $\hat{A}\Psi_i = a_i\Psi_i$ for an operator $\hat{A}$ corresponding to an observable A , then the eigenvalue a_i is the result of measuring A for the state described by Ψ_i . Here observable A is well defined for this eigenstate.

For n such states, there are an associated set of eigenvalues, a_i ($i = 1, 2, \dots, n$). A measurement associated with observable A could return any of these eigenvalues.

Motivated by our understanding of classical mechanics where measurements of physical observables, e.g. position and momentum, are represented by real, rather than complex, values, then we will assert that the eigenvalues a_i are real.

Operators that produce real eigenvalues are called **Hermitian** (all QM operators are found to be Hermitian). Hermitian operators satisfy

$$\int_{-\infty}^{\infty} \Psi_b^* \hat{O} \Psi_a dx = \int_{-\infty}^{\infty} \Psi_a (\hat{O} \Psi_b)^* dx$$

Demonstration. Let Ψ_a be an eigenfunction of $\hat{O}$ such that $\hat{O}\Psi_a = a\Psi_a$.

If $\hat{O}$ is Hermitian, then

$$\int_{-\infty}^{\infty} \Psi_a^* \hat{O} \Psi_a dx = \int_{-\infty}^{\infty} \Psi_a (\hat{O} \Psi_a)^* dx$$

$$\int_{-\infty}^{\infty} \Psi_a^* a \Psi_a dx = \int_{-\infty}^{\infty} \Psi_a (a \Psi_a)^* dx$$

$$a \int_{-\infty}^{\infty} \Psi_a^* \Psi_a dx = a^* \int_{-\infty}^{\infty} \Psi_a \Psi_a^* dx$$

If the wavefunctions are normalised then

$$\int_{-\infty}^{\infty} \Psi_a^* \Psi_a dx = \int_{-\infty}^{\infty} \Psi_a \Psi_a^* dx = 1$$

And $a = a^*$, which is only possible if the eigenvalues are real.

Relationship Between Eigenfunctions

Let Ψ_b be another eigenfunction of $\hat{O}$ such that $\hat{O} \Psi_b = b \Psi_b$.

$$\int_{-\infty}^{\infty} \Psi_b^* \hat{O} \Psi_a dx = \int_{-\infty}^{\infty} \Psi_b (\hat{O} \Psi_a)^* dx$$

$$a \int_{-\infty}^{\infty} \Psi_b^* \Psi_a dx = b^* \int_{-\infty}^{\infty} \Psi_a (\Psi_b)^* dx = b \int_{-\infty}^{\infty} \Psi_a (\Psi_b)^* dx = b \int_{-\infty}^{\infty} \Psi_b^* \Psi_a dx$$

So

$$(a - b) \int_{-\infty}^{\infty} \Psi_b^* \Psi_a dx = 0$$

There are two solutions

$$i) a \neq b \rightarrow \int_{-\infty}^{\infty} \Psi_b^* \Psi_a dx = 0$$

$$ii) a = b \rightarrow \int_{-\infty}^{\infty} \Psi_a^* \Psi_a dx = 1$$

This can be combined into a single equation

$$\int_{-\infty}^{\infty} \Psi_b^* \Psi_a dx = \delta_{ab}$$

Where δ_{ab} is the Kronecker delta ($\delta_{ab} = 1, a = b$; $\delta_{ab} = 0, a \neq b$).

So the eigenfunctions of a Hermitian operator are **orthogonal** (as are the principal axes of the Cartesian coordinate system), or **orthonormal**.

Expectation Values

What are the expectation values of a superposition state such as

$$\Phi(x, t) = \sum_{i=1}^n c_i \Psi_i(x, t)$$

Assuming $\Phi(x, t)$ is normalised.

$$\langle A \rangle = \int_{-\infty}^{\infty} \Phi^* \hat{A} \Phi dx$$

$$\langle A \rangle = \int_{-\infty}^{\infty} \left[\sum_{i=1}^n c_i \Psi_i(x, t) \right]^* \hat{A} \left[\sum_{j=1}^n c_j \Psi_j(x, t) \right] dx$$

Where i, j are independent.

$$\langle A \rangle = \int_{-\infty}^{\infty} \left[\sum_{i=1}^n c_i \Psi_i \right]^* \left[\sum_{j=1}^n c_j a_j \Psi_j \right] dx$$

$$\langle A \rangle = \sum_{i=1}^n \sum_{j=1}^n c_i^* c_j a_j \int_{-\infty}^{\infty} \Psi_i^* \Psi_j dx$$

$$\langle A \rangle = \sum_{i=1}^n \sum_{j=1}^n c_i^* c_j a_j \delta_{ij}$$

$$\langle A \rangle = \sum_{i=1}^n c_i^* c_i a_i = \sum_{i=1}^n |c_i|^2 a_i$$

The expectation value is the weighted mean of all possible values of the observable; $|c_j|^2$ is the probability of measuring a_j .

In general:

$$\langle A \rangle = \sum_{i=1}^n a_i |c_i|^2$$

Also because

$$\int_{-\infty}^{\infty} \Phi^* \Phi dx = \sum_{i=1}^n |c_i|^2$$

And the wave-functions are normalised, then so are the probabilities (they sum to unity)

$$\sum_{i=1}^n |c_i|^2 = 1$$

$$\begin{aligned} \int_{-\infty}^{\infty} \Phi^* \Phi dx &= \int_{-\infty}^{\infty} \sum_{i=1}^n c_i^* \Psi_i^*(x, t) \sum_{j=1}^n c_j \Psi_j(x, t) dx \\ &= \int_{-\infty}^{\infty} \sum_{i=1, j=1}^{n, n} c_i^* c_j \Psi_i^*(x, t) \Psi_j(x, t) dx \\ &= \sum_{i=1, j=1}^{n, n} c_i^* c_j \int_{-\infty}^{\infty} \Psi_i^*(x, t) \Psi_j(x, t) dx = \sum_{i=1, j=1}^{n, n} c_i^* c_j \delta_{ij} \\ \int_{-\infty}^{\infty} \Phi^* \Phi dx &= \sum_{i=1}^n |c_i|^2 \end{aligned}$$

Example:

Particle in an infinite square well.

The ground-state wave-function is

$$\underbrace{\phi(x)}_{\text{superposition}} = \sqrt{\frac{2}{a}} \cos\left(\frac{\pi x}{a}\right)$$

$\phi(x)$ is an eigenfunction of KE

$$\hat{T} \phi(x) = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \sqrt{\frac{2}{a}} \cos\left(\frac{\pi x}{a}\right) = \underbrace{\frac{\hbar^2}{2m} \left(\frac{\pi}{a}\right)^2}_E \phi(x)$$

So $\hat{T}$ is well defined for this state.

However, $\phi(x)$ is not an eigenfunction of momentum.

$$\hat{p} \phi(x) = -i\hbar \frac{d}{dx} \sqrt{\frac{2}{a}} \cos\left(\frac{\pi x}{a}\right) = \frac{i\hbar\pi}{a} \sqrt{\frac{2}{a}} \sin\left(\frac{\pi x}{a}\right) \neq \text{const} \times \phi(x)$$

We can think about $\phi(x)$ as a superposition of two eigenstates of momentum (two waves travelling in opposite directions), in this case a complete set of two possibilities.

The eigenfunctions of momentum are

$$\begin{aligned} \psi_+(x) &= \frac{1}{\sqrt{a}} e^{ikx}; & k &= \frac{\pi}{a}; & \hat{p}\psi_+ &= +\frac{\hbar\pi}{a} \psi_+ \\ \psi_-(x) &= \frac{1}{\sqrt{a}} e^{-ikx}; & k &= \frac{\pi}{a}; & \hat{p}\psi_- &= -\frac{\hbar\pi}{a} \psi_- \end{aligned}$$

Normalisation:

$$\int_{-a/2}^{a/2} \psi_+^*(x) \psi_+(x) dx = \int_{-a/2}^{a/2} \psi_-^*(x) \psi_-(x) dx = \frac{1}{a} [x]_{-a/2}^{a/2} = 1$$

It is possible to write $\phi(x)$ in terms of this complete basis in $\psi_{\pm}(x)$.

$$\phi(x) = \sqrt{\frac{2}{a}} \left[\frac{e^{\frac{i\pi}{a}x} + e^{-\frac{i\pi}{a}x}}{2} \right]$$

$$\phi(x) = \frac{1}{\sqrt{2}} \psi_+(x) + \frac{1}{\sqrt{2}} \psi_-(x) \equiv c_+ \psi_+(x) + c_- \psi_-(x)$$

$$c_{\pm} = \frac{1}{\sqrt{2}}$$

Probability of measuring $p_{\pm} = |c_{\pm}|^2 = 1/2$.

$$\langle p \rangle = \frac{1}{2} \left(+\frac{\hbar\pi}{a} \right) + \frac{1}{2} \left(-\frac{\hbar\pi}{a} \right) = 0$$

In general, we can use the orthogonality of the eigenfunctions to calculate the coefficients.

$$\phi(x) = \sum_{i=1}^n c_i \psi_i(x)$$

With both $\phi(x)$ and $\psi_i(x)$ normalised. Then

$$\underbrace{\int_{-\infty}^{\infty} \psi_j(x)^* \phi(x) dx}_{\text{overlap integral (inner product)}} = \sum_{i=1}^n c_i \int_{-\infty}^{\infty} \psi_j(x)^* \psi_i(x) dx = \sum_{i=1}^n c_i \delta_{ij} = c_j$$

$$\int_{-\infty}^{\infty} \psi_j(x)^* \phi(x) dx = c_j$$

Continuing with the example.....

$$c_+ = \int_{-a/2}^{a/2} \psi_+^*(x) \phi(x) dx$$

$$c_+ = \int_{-a/2}^{a/2} \left(\frac{1}{\sqrt{a}} e^{-ikx} \right) \left(\sqrt{\frac{2}{a}} \cos\left(\frac{\pi}{a}x\right) \right) dx$$

$$c_+ = \frac{\sqrt{2}}{a} \int_{-a/2}^{a/2} \left(\cos(kx) - i \underbrace{\sin(kx)}_{\text{odd function}=0} \right) \cos\left(\frac{\pi}{a}x\right) dx$$

$$c_+ = \frac{\sqrt{2}}{a} \int_{-\frac{a}{2}}^{\frac{a}{2}} \frac{1}{2} (1 + \cos(2kx)) dx = \frac{1}{a\sqrt{2}} \left[x + \frac{1}{2k} \underbrace{\sin(2kx)}_{=0} \right]_{-\frac{a}{2}}^{\frac{a}{2}} ; \quad k = \frac{\pi}{a}$$

$$c_+ = \frac{1}{\sqrt{2}}$$

Summary

- We have been considering a superposition state $\Phi(x, t) = \sum_{i=1}^n c_i \Psi_i(x, t)$ where $\Psi_i(x, t)$ is an eigenfunction of an operator $\hat{A}$ such that $\hat{A}\Psi_i = a_i \Psi_i$.
- Before measurement the value of A is indeterminate (not just unknown)
- When A is measured, the result will be a_i with a probability $|c_i|^2$.
- After measurement the wave-function is Ψ_i
- All knowledge of the earlier state Φ is lost.

Postulate 4: The result of measurement on a superposition state is that the wave-function **collapses** onto one of the eigenstates corresponding to the observable being measured. This is the Copenhagen Interpretation

2. Simultaneous Eigenfunctions

A wavefunction can sometimes be an eigenfunction of two (or more) operators.

For example, $\Psi(x, t) = Ae^{i(kx - \omega t)}$

$$\hat{p} = -i\hbar \frac{\partial}{\partial x} ; \quad \hat{p}\Psi = p\Psi; \quad p = \hbar k$$

$$\hat{E} = i\hbar \frac{\partial}{\partial t} ; \quad \hat{E}\Psi = E\Psi; \quad E = \hbar\omega$$

This happens when the operators **commute**. The order of operations is not important – doesn't matter if you measure E or p first as the wave-function is unchanged after the first measurement.

In general, operators do not commute, e.g.

$$\underbrace{x}_{\text{multiply}} \cdot \underbrace{\frac{d}{dx}}_{\text{differentiate}} [f(x)] \neq \frac{d}{dx} [x \cdot f(x)]$$

$$x \cdot \frac{df}{dx} \neq f(x) + x \frac{df}{dx}$$

Consider two operators $\hat{A}$ and $\hat{B}$

$$\hat{A}(\hat{B}\Psi) \neq \hat{B}(\hat{A}\Psi) \quad \text{in general}$$

$$\underbrace{(\hat{A}\hat{B} - \hat{B}\hat{A})}_{[\hat{A}, \hat{B}]} \Psi \neq 0$$

$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$ is called the **commutator** of $\hat{A}$ and $\hat{B}$.

If $[\hat{A}, \hat{B}]\Psi \neq 0$ $[\hat{A}, \hat{B}] \neq 0$ $\hat{A}$ and $\hat{B}$ do not commute.

If $[\hat{A}, \hat{B}]\Psi = 0$ $[\hat{A}, \hat{B}] = 0$ $\hat{A}$ and $\hat{B}$ commute.

Can we show that $\hat{A}$ and $\hat{B}$ commute if they share a complete set of eigenfunctions?

Let, $\hat{A}\Psi_i = a_i\Psi_i$ and $\hat{B}\Psi_i = b_i\Psi_i$

Then $[\hat{A}, \hat{B}]\Psi_i = \hat{A}\hat{B}\Psi_i - \hat{B}\hat{A}\Psi_i$

$$[\hat{A}, \hat{B}]\Psi_i = \hat{A}b_i\Psi_i - \hat{B}a_i\Psi_i = b_i\hat{A}\Psi_i - a_i\hat{B}\Psi_i = b_ia_i\Psi_i - a_ib_i\Psi_i = 0$$

Since b_i and a_i are real then $b_ia_i = a_ib_i$.

The above is true for any value of i so it should also be true for any superposition state $\Phi(x, t) = \sum_{i=1}^n c_i\Psi_i(x, t)$, since the commutator is formed from two linear operators:

$$[\hat{A}, \hat{B}]\Phi = [\hat{A}, \hat{B}] \sum_{i=1}^n c_i\Psi_i(x, t) = \sum_{i=1}^n c_i[\hat{A}, \hat{B}]\Psi_i(x, t) = 0$$

What does this mean?

Consider what happens when A and then B are measured for the superposition state Φ .

1. Measure A , result is a_i with probability $|c_i|^2$. Φ collapses to state Ψ_i .
2. Now measure B , result is b_i with 100% since $\hat{B}\Psi_i = b_i\Psi_i$.
3. If A and B commute, then it is possible to know A and B simultaneously, without any uncertainty.
4. If not, i.e. A and B don't commute, then A and B cannot be known at the same time.

It is also possible to show that if $\hat{A}$ and $\hat{B}$ commute, they share a complete set of eigenfunctions (see the notes on Canvas).

Also Canvas has some proofs of commutator relations.

3. Operators that Don't Commute and Uncertainty

Consider two operators that don't commute ($\Delta x \Delta p \geq \hbar/2$)

$$[\hat{x}, \hat{p}]\Psi(x, t) = \left[x, -i\hbar \frac{\partial}{\partial x} \right] \Psi(x, t) = -i\hbar \left(x \frac{\partial \Psi}{\partial x} - \frac{\partial}{\partial x} (x \Psi) \right)$$

$$[\hat{x}, \hat{p}]\Psi(x, t) = -i\hbar \left(x \frac{\partial \Psi}{\partial x} - \Psi - x \frac{\partial \Psi}{\partial x} \right) = i\hbar \Psi$$

Hence

$$[\hat{x}, \hat{p}] = i\hbar$$

The measurement of position and momentum do not commute, as the eigenfunctions of position and momentum are different. So cannot know the precise values of x and p_x simultaneously.

How does this lead to the uncertainty principle?

Uncertainty Principle and Fourier Transforms

Remembering

$$\phi(x) = \sum_{i=1}^n c_i \psi_i(x)$$

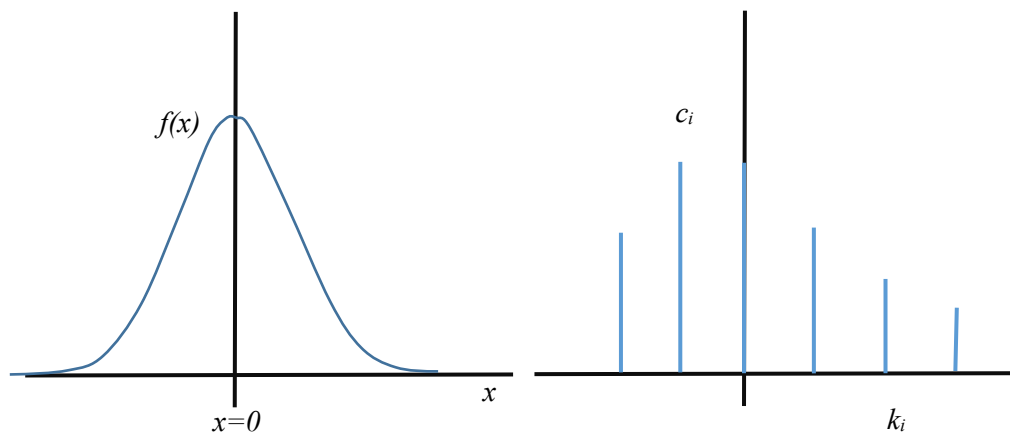
Consider we have a complete set of momentum states given by

$$\psi_p(x) = e^{ikx}; k = \frac{p}{\hbar}$$

Then we can write any arbitrary function in terms of these basis states:

$$f(x) = \sum_{i=1}^n c_i e^{ik_i x}, \text{ where}$$

$$c_i = \int_{-\infty}^{\infty} \psi_p^* f(x) dx = \int_{-\infty}^{\infty} e^{-ik_i x} f(x) dx$$



If k_i is continuous then we can write

$$f(x) = \int_{-\infty}^{\infty} c(k) e^{ik_i x} dk$$

$$c(k) = \int_{-\infty}^{\infty} f(x) e^{-ik_i x} dx = \phi(k) \quad [\text{wavefunction in momentum space}]$$

These are then Fourier Transforms between $f(x)$ and $c(k)$.

If

$$f(x) = e^{-x^2/2a^2} \quad [\text{Gaussian with width set by } a]$$

Then the probability of finding a particle at a particular location is given by

$$P(x) = |f(x)|^2 = e^{-x^2/a^2}$$

$$\phi(k) = \int_{-\infty}^{\infty} e^{-x^2/2a^2} e^{-ikx} dx$$

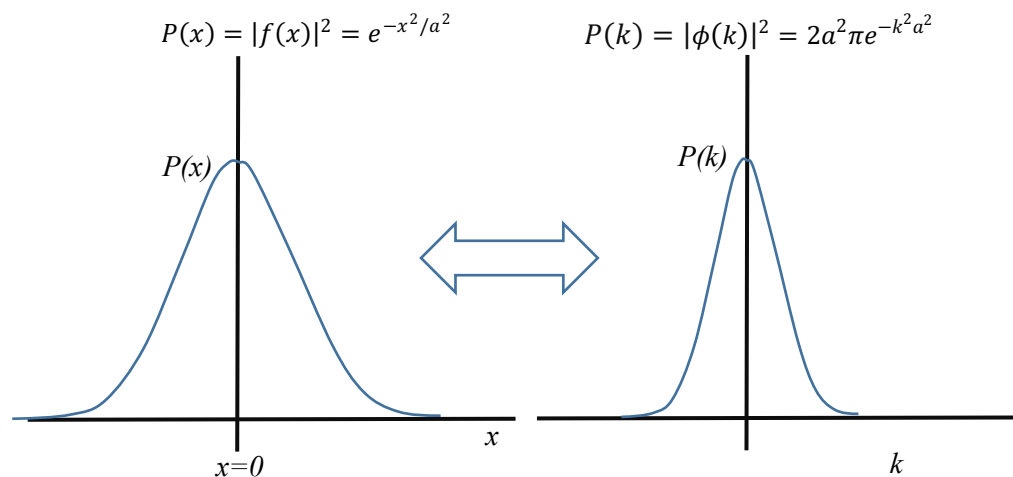
$$\phi(k) = \int_{-\infty}^{\infty} \exp\left\{-\frac{1}{2}\left[\left(\frac{x}{a}\right)^2 + 2ikx\right]\right\} dx$$

$$\phi(k) = \int_{-\infty}^{\infty} \exp\left\{-\frac{1}{2}\left[\left(\frac{x}{a} + ika\right)^2 + k^2a^2\right]\right\} dx$$

$$\phi(k) = e^{-k^2a^2/2} \int_{-\infty}^{\infty} \exp\left\{-\frac{1}{2}\left[\left(\frac{x}{a} + ika\right)^2\right]\right\} dx$$

$$\phi(k) = \sqrt{2a^2\pi} e^{-k^2a^2/2}$$

$$\text{Hence } P(k) = |\phi(k)|^2 = 2a^2\pi e^{-k^2a^2}$$



The momentum wavefunction is the Fourier Transform of the spatial wavefunction and the opposite is true.

Uncertainty relations exist for all non-commuting pairs of operators.

For example there is a similar relationship for energy and time.

$$\Delta E \Delta t \geq \frac{\hbar}{2}$$

Which follows from $[\hat{E}, \hat{t}]\Psi(x, t) = i\hbar\Psi$

Note: When considering the Heisenberg Uncertainty Principle imagine that we create multiple copies of a system each described by the same wavefunction. If we carry out a set of position measurements on some and momentum measurements on the others then the product of the dispersions of the measurements will never be less than $\hbar/2$.

This principle does not apply to sequential measurements of position and momentum. For two operators that do not commute the first measurement collapses the wave-function on to one of the corresponding eigenfunctions of the first operator, not shared by the second. This actually leads to a different lower bound for sequential measurements of two non-commuting operators on the same system; $\Delta x \Delta p \geq \hbar$.

4. The Schrödinger Equation in 3-D

In Cartesian coordinates

$$\hat{\underline{p}} = (\hat{p}_x, \hat{p}_y, \hat{p}_z) = -i\hbar \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right) = -i\hbar \underline{\nabla}$$

Where $\underline{\nabla}$ is called nabla and here has a vector form.

$$\hat{T} = \frac{\hat{\underline{p}}^2}{2m} = -\frac{\hbar^2}{2m} \nabla^2$$

∇^2 is called the Laplacian

$$\nabla^2 = \underline{\nabla} \cdot \underline{\nabla} = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right)$$

In 3-D the SE becomes

$$-\frac{\hbar^2}{2m} \nabla^2 \Psi(\underline{r}, t) + V(\underline{r}, t) \Psi(\underline{r}, t) = i\hbar \frac{\partial}{\partial t} \Psi(\underline{r}, t)$$

If $V(\underline{r}, t) \rightarrow V(\underline{r})$ then the SE becomes the TISE.

$$-\frac{\hbar^2}{2m} \nabla^2 \psi(\underline{r}) + V(\underline{r}) \psi(\underline{r}) = E \psi(\underline{r})$$

With $\Psi(\underline{r}, t) = \psi(\underline{r}) e^{-\frac{iEt}{\hbar}}$ as before.

The travelling wave in 3-D is given by

$$\psi(\underline{r}) = A e^{i(\underline{k} \cdot \underline{r})}$$

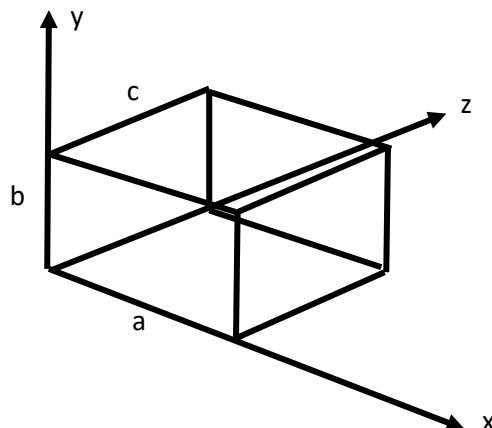
Where $\underline{k} = (k_x, k_y, k_z)$ is the wave vector.

As before $\psi(\underline{r})$ is an eigenfunction of $\hat{\underline{p}}$.

$$\hat{\underline{p}} e^{i\underline{k} \cdot \underline{r}} = -i\hbar \underline{\nabla} e^{i\underline{k} \cdot \underline{r}} = \hbar \underline{k} e^{i\underline{k} \cdot \underline{r}} = \underline{p} e^{i\underline{k} \cdot \underline{r}}$$

Note i) that the uncertainty relations apply in each direction e.g. $[\hat{y}, \hat{p}_y] = i\hbar$ and $\Delta y \cdot \Delta p_y \geq \hbar/2$
and ii) the position and momentum operators in perpendicular directions do commute, e.g. $[\hat{x}, \hat{p}_y] = 0$. So it is possible to know precisely the values of x and p_y simultaneously.

Consider solutions of the TISE in 3D



Let $V(\underline{r}) = 0$ for

$$0 \leq x \leq a$$

$$0 \leq y \leq b$$

$$0 \leq z \leq c$$

and $V(\underline{r}) = \infty$ elsewhere.

Outside the box: $V(\underline{r}) = \infty$ and $\psi(\underline{r}) = 0$

Inside the box:

$$-\frac{\hbar^2}{2m} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \psi(\underline{r}) = E\psi(\underline{r})$$

Solving using separation of variables.....

Let $\psi(\underline{r}) = X(x)Y(y)Z(z)$

$$-\frac{\hbar^2}{2m} \left(Y(y)Z(z) \frac{\partial^2 X(x)}{\partial x^2} + X(x)Z(z) \frac{\partial^2 Y(y)}{\partial y^2} + X(x)Y(y) \frac{\partial^2 Z(z)}{\partial z^2} \right) = EX(x)Y(y)Z(z)$$

Divide through by $\psi(\underline{r})$

$$-\frac{\hbar^2}{2m} \left(\frac{1}{X(x)} \frac{\partial^2 X(x)}{\partial x^2} + \frac{1}{Y(y)} \frac{\partial^2 Y(y)}{\partial y^2} + \frac{1}{Z(z)} \frac{\partial^2 Z(z)}{\partial z^2} \right) = E$$

Each term on the LHS is independent and this must equal a different constant, so set

$$E = E_x + E_y + E_z$$

So then we have 3 ODEs e.g.

$$-\frac{\hbar^2}{2m} \frac{\partial^2 X(x)}{\partial x^2} = E_x X(x)$$

$$-\frac{\hbar^2}{2m} \frac{\partial^2 Y(y)}{\partial y^2} = E_y Y(y)$$

$$-\frac{\hbar^2}{2m} \frac{\partial^2 Z(z)}{\partial z^2} = E_z Z(z)$$

Where the solutions are

$$X(x) = A_x \sin\left(\frac{n_x \pi x}{a}\right)$$

$$n_x = 1, 2, 3 \dots$$

$$Y(y) = A_y \sin\left(\frac{n_y \pi y}{b}\right)$$

$$n_y = 1, 2, 3 \dots$$

$$Z(z) = A_z \sin\left(\frac{n_z \pi z}{c}\right)$$

$$n_z = 1, 2, 3 \dots$$

[Only have sine solutions as corner of the box is at the origin in the coordinate system.]

Combining the solutions we have :

$$\psi(r) = A \sin\left(\frac{n_x \pi x}{a}\right) \sin\left(\frac{n_y \pi y}{b}\right) \sin\left(\frac{n_z \pi z}{c}\right)$$

With $k_x = n_x \pi / a$, $k_y = n_y \pi / b$ and $k_z = n_z \pi / c$.

Combining the eigenvalues

$$E = E_x + E_y + E_z$$

$$E = \frac{\hbar^2 \pi^2}{2m} \left(\frac{n_x^2}{a^2} + \frac{n_y^2}{b^2} + \frac{n_z^2}{c^2} \right)$$

With $n_x, n_y, n_z = 1, 2, 3 \dots$ (all independently)

Example:

Let $a = b = c$ (cubic potential)

$$E = \frac{\hbar^2 \pi^2}{2ma^2} (n_x^2 + n_y^2 + n_z^2)$$

What are the energy levels

State	(n_x, n_y, n_z)	E	Degeneracy
Ground State	(1,1,1)	$3 \frac{\hbar^2 \pi^2}{2ma^2}$	1
1 st Excited State	(2,1,1)	$6 \frac{\hbar^2 \pi^2}{2ma^2}$	3
	(1,2,1)		
	(1,1,2)		

2 nd Excited State	(2,2,1)	$9 \frac{\hbar^2 \pi^2}{2ma^2}$	3
	(2,1,2)		
	(1,2,2)		
3 rd Excited State	(3,1,1)	$11 \frac{\hbar^2 \pi^2}{2ma^2}$	3
	(1,3,1)		
	(1,1,3)		
4 th Excited State	(2,2,2)	$12 \frac{\hbar^2 \pi^2}{2ma^2}$	1

Note that different combinations of n_x, n_y, n_z can give the same energy eigenvalue and then these levels are said to be *degenerate*. This arises due to a symmetry of the potential.

The number of states with the same energy is the *degeneracy value*. The degeneracy can be removed if $a \neq b \neq c$.

5. Angular Momentum

Once one has three dimensions, one can have motion in orbits and hence angular momentum. This will feature in both atomic and nuclear systems.

Classically

$$\underline{L} = \underline{r} \times \underline{p}$$

In matrix (determinant) form

$$\underline{L} = \begin{pmatrix} \hat{i} & \hat{j} & \hat{k} \\ x & y & z \\ p_x & p_y & p_z \end{pmatrix}$$

This gives us the components:

$$L_x = yp_z - zp_y; L_y = zp_x - xp_z; L_z = xp_y - yp_x$$

$$\begin{aligned} \hat{L}_x &= -i\hbar \left(y \frac{\partial}{\partial z} - z \frac{\partial}{\partial y} \right) \\ \hat{L}_y &= -i\hbar \left(z \frac{\partial}{\partial x} - x \frac{\partial}{\partial z} \right) \\ \hat{L}_z &= -i\hbar \left(x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x} \right) \end{aligned}$$

Can two components be known simultaneously?

$$[\hat{L}_x, \hat{L}_y] = \hat{L}_x \hat{L}_y - \hat{L}_y \hat{L}_x$$

$$\begin{aligned}
[\hat{L}_x, \hat{L}_y] &= -\hbar^2 \left\{ \left(y \frac{\partial}{\partial z} - z \frac{\partial}{\partial y} \right) \left(z \frac{\partial}{\partial x} - x \frac{\partial}{\partial z} \right) - \left(z \frac{\partial}{\partial x} - x \frac{\partial}{\partial z} \right) \left(y \frac{\partial}{\partial z} - z \frac{\partial}{\partial y} \right) \right\} \\
[\hat{L}_x, \hat{L}_y] &= -\hbar^2 \left\{ \left(y \frac{\partial}{\partial z} z \frac{\partial}{\partial x} - z \frac{\partial}{\partial y} z \frac{\partial}{\partial x} \right) - \left(y \frac{\partial}{\partial z} x \frac{\partial}{\partial z} - z \frac{\partial}{\partial y} x \frac{\partial}{\partial z} \right) - \left(z \frac{\partial}{\partial x} y \frac{\partial}{\partial z} - x \frac{\partial}{\partial z} y \frac{\partial}{\partial z} \right) \right. \\
&\quad \left. + \left(z \frac{\partial}{\partial x} z \frac{\partial}{\partial y} - x \frac{\partial}{\partial z} z \frac{\partial}{\partial y} \right) \right\} \\
[\hat{L}_x, \hat{L}_y] &= -\hbar^2 \left\{ y \frac{\partial}{\partial x} + yz \frac{\partial^2}{\partial z \partial x} - z^2 \frac{\partial^2}{\partial x \partial y} - yx \frac{\partial^2}{\partial z^2} + zx \frac{\partial^2}{\partial y \partial z} - zy \frac{\partial^2}{\partial x \partial z} + xy \frac{\partial^2}{\partial z^2} + z^2 \frac{\partial^2}{\partial x \partial y} \right. \\
&\quad \left. - xz \frac{\partial^2}{\partial z \partial y} - x \frac{\partial}{\partial y} \right\} \\
[\hat{L}_x, \hat{L}_y] &= -\hbar^2 \left\{ \left(y \frac{\partial}{\partial x} - 0 \right) - (0 - 0) - (0 - 0) + \left(0 - x \frac{\partial}{\partial y} \right) \right\} \\
[\hat{L}_x, \hat{L}_y] &= \hbar^2 \left(x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x} \right) = i\hbar \hat{L}_z
\end{aligned}$$

So

$$\begin{aligned}
[\hat{L}_x, \hat{L}_y] &= i\hbar \hat{L}_z \\
[\hat{L}_y, \hat{L}_z] &= i\hbar \hat{L}_x \\
[\hat{L}_z, \hat{L}_x] &= i\hbar \hat{L}_y
\end{aligned}$$

No two components commute and hence can only know one component of angular momentum precisely at a time and we cannot know the angular momentum completely.

However, consider the magnitude of $\underline{L}$.

$\underline{L} = (\hat{L}_x, \hat{L}_y, \hat{L}_z)$: This is a vector

$\hat{L}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2$: This is a scalar

$$[\hat{L}^2, \hat{L}_z] = [\hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2, \hat{L}_z]$$

$$[\hat{L}^2, \hat{L}_z] = [\hat{L}_x^2, \hat{L}_z] + [\hat{L}_y^2, \hat{L}_z] + [\hat{L}_z^2, \hat{L}_z]$$

$$\begin{aligned}
[\hat{L}_z^2, \hat{L}_z] &= i\hbar^3 \left[\left(x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x} \right) \left(x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x} \right) \left(x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x} \right) \right. \\
&\quad \left. - \left(x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x} \right) \left(x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x} \right) \left(x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x} \right) \right] = 0
\end{aligned}$$

Also noting in general

$$[\hat{A}\hat{B}, \hat{C}] = \hat{A}\hat{B}\hat{C} - \hat{C}\hat{A}\hat{B}$$

$$[\hat{A}\hat{B}, \hat{C}] = \hat{A}\hat{B}\hat{C} - \underbrace{\hat{A}\hat{C}\hat{B} + \hat{A}\hat{C}\hat{B}}_{=0} - \hat{C}\hat{A}\hat{B} = \hat{A}[\hat{B}, \hat{C}] + [\hat{A}, \hat{C}]\hat{B}$$

$$\text{and } [\hat{A}, \hat{B}] = -[\hat{B}, \hat{A}]$$

Then

$$[\hat{L}_x^2, \hat{L}_z] = [\hat{L}_x \hat{L}_x, \hat{L}_z] = \hat{L}_x [\hat{L}_x, \hat{L}_z] + [\hat{L}_x, \hat{L}_z] \hat{L}_x$$

$$[\hat{L}_x^2, \hat{L}_z] = \hat{L}_x (-i\hbar \hat{L}_y) + (-i\hbar \hat{L}_y) \hat{L}_x = -i\hbar (\hat{L}_x \hat{L}_y + \hat{L}_y \hat{L}_x)$$

Similarly

$$[\hat{L}_y^2, \hat{L}_z] = [\hat{L}_y \hat{L}_y, \hat{L}_z] = \hat{L}_y [\hat{L}_y, \hat{L}_z] + [\hat{L}_y, \hat{L}_z] \hat{L}_y$$

$$[\hat{L}_y^2, \hat{L}_z] = \hat{L}_y (i\hbar \hat{L}_x) + (i\hbar \hat{L}_x) \hat{L}_y = i\hbar (\hat{L}_x \hat{L}_y + \hat{L}_y \hat{L}_x)$$

Hence

$$[\hat{L}^2, \hat{L}_z] = 0$$

One could also repeat this with the x and y components.

So

$$[\hat{L}^2, \hat{L}_x] = 0$$

and

$$[\hat{L}^2, \hat{L}_y] = 0$$

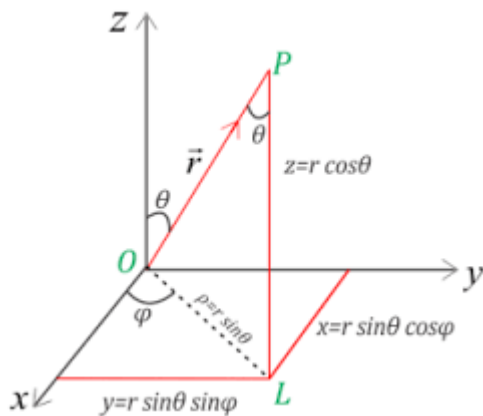
So we can know the precise values of $\hat{L}^2$ and one component of $\hat{L}$ only (by convention we choose z).

We can know the magnitude of $\hat{L}$, but not its direction.

Central Potentials

Potential energy varies only with distance (not angle) from the origin (e.g. the monopole Coulomb potential).

$$V(x, y, z) \rightarrow V(r, \theta, \phi) \rightarrow V(r)$$



Angular Momentum in Spherical Polar Coordinates

We would like to express angular momentum operators in spherical polar coordinates rather than Cartesian, as it provides a better description of orbital processes.

$$\hat{L}_z = -i\hbar \left(x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x} \right) = -i\hbar \frac{\partial}{\partial \phi}$$

Proof

$$\frac{\partial}{\partial \phi} = \frac{\partial}{\partial x} \frac{\partial x}{\partial \phi} + \frac{\partial}{\partial y} \frac{\partial y}{\partial \phi} + \frac{\partial}{\partial z} \frac{\partial z}{\partial \phi}$$

Using

$$x = r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta$$

$$\frac{\partial x}{\partial \phi} = -r \sin \theta \sin \phi = -y$$

$$\frac{\partial y}{\partial \phi} = r \sin \theta \cos \phi = x$$

$$\frac{\partial z}{\partial \phi} = 0$$

So.....

$$\frac{\partial}{\partial \phi} = -y \frac{\partial}{\partial x} + x \frac{\partial}{\partial y}$$

So

$$\hat{L}_z = -i\hbar \frac{\partial}{\partial \phi}$$

Other operators:

$$\hat{L}_x = i\hbar \left(\sin \phi \frac{\partial}{\partial \theta} + \cot \theta \cos \phi \frac{\partial}{\partial \phi} \right)$$

$$\hat{L}_y = i\hbar \left(-\cos \phi \frac{\partial}{\partial \theta} + \cot \theta \sin \phi \frac{\partial}{\partial \phi} \right)$$

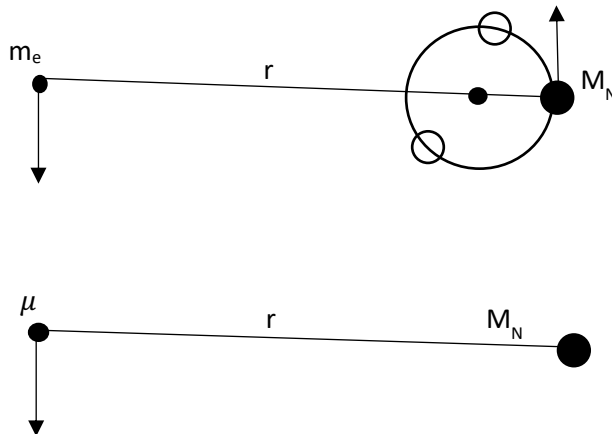
$$\hat{L}^2 = -\hbar^2 \left(\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right)$$

Single Electron Atoms (see Eisberg and Resnick, Chapter 7 and Appendix N for details)

Consider the TISE with a Coulomb potential, with an electron orbiting a nucleus....

$$V(r) = -\frac{Ze^2}{4\pi\epsilon_0 r}$$

Classically the electron and the atomic nucleus rotate around a common centre-of-mass.



The first picture illustrates the actual case, where the electron and nucleus orbit around the centre-of-mass and in this case the origin of the potential is not fixed. The mass of the electron is replaced by the reduced mass. However, as the nucleus is much more massive than the electron (by about a factor of 2000) the radius of the orbit of the nucleus is very small and the situation is more like the second picture, where the electron orbits a quasi-stationary nucleus.

The reduced mass of the electron is given by:

$$\mu = \left(\frac{m_N m_e}{m_N + m_e} \right)$$

The TISE becomes

$$\left(-\frac{\hbar^2}{2\mu} \nabla^2 + V(r) \right) \psi(r, \theta, \phi) = E \psi(r, \theta, \phi)$$

Here we have spherical polar coordinates and (see Canvas) hence the Laplacian becomes

$$\nabla^2 = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2}{\partial \phi^2}$$

Solve TISE using separation of variables:

$$\psi(r, \theta, \phi) = R(r)\Theta(\theta)\Phi(\phi)$$

Substituting and carrying out the partial differentiations, we obtain

$$-\frac{\hbar^2}{2\mu} \left[\frac{\Theta \Phi}{r^2} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \frac{R \Phi}{r^2 \sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) + \frac{R \Theta}{r^2 \sin^2 \theta} \frac{d^2 \Phi}{d\phi^2} \right] + V(r) R \Theta \Phi = E R \Theta \Phi$$

To obtain the azimuthal equation, multiply by

$$-\frac{2\mu r^2 \sin^2 \theta}{R \Theta \hbar^2}$$

and rearrange. This gives

$$\frac{1}{\Phi} \frac{d^2 \Phi}{d\phi^2} = -\frac{\sin^2 \theta}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) - \frac{\sin \theta}{\Theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) - \frac{2\mu}{\hbar^2} r^2 \sin^2 \theta (E - V(r)).$$

As the left hand side depends only on ϕ and the right hand side depends on r and θ , this mathematical statement can only be true if both sides are equal to a constant. This constant is chosen to be $-m^2$. This results in two equations:

$$\frac{d^2 \Phi}{d\phi^2} = -m^2 \Phi$$

and

$$-\frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) - \frac{1}{\Theta \sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) - \frac{2\mu}{\hbar^2} r^2 (E - V(r)) = -\frac{m^2}{\sin^2 \theta}.$$

Gathering the terms that only depend on r on the left hand side and θ on the right hand side, and multiplying through by -1 , we obtain

$$\frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \frac{2\mu}{\hbar^2} r^2 (E - V(r)) = \frac{m^2}{\sin^2 \theta} - \frac{1}{\Theta \sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right).$$

Once again, the terms on the left hand side of the equation are independent of those on the right, so we set both sides equal to a constant. This time, the constant is chosen to be $l(l+1)$. Thus, we end up with two more equations: the polar equation

$$-\frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) + \frac{m^2 \Theta}{\sin^2 \theta} = l(l+1) \Theta$$

and the radial equation

$$\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \frac{2\mu}{\hbar^2} (E - V(r)) R = l(l+1) \frac{R}{r^2}.$$

Azimuthal Dependence

Taking the azimuthal equation

$$\frac{d^2\Phi}{d^2\phi} = -m^2\Phi$$

Solutions have the form (can check by substitution)

$$\Phi(\phi) = Ae^{im\phi}$$

This is normalised such that

$$\int_0^{2\pi} \Phi^* \Phi d\phi = 1$$

$$\int_0^{2\pi} Ae^{-im\phi} Ae^{im\phi} d\phi = A^2 2\pi$$

Hence

$$A = 1/\sqrt{2\pi}$$

$\Phi(\phi) = Ae^{im\phi}$ is an eigenfunction of $\hat{L}_z$ (which corresponds to a rotation about the z-axis).

$$\hat{L}_z \Phi(\phi) = -i\hbar \frac{\partial}{\partial \phi} \left(\frac{1}{\sqrt{2\pi}} e^{im\phi} \right) = m\hbar \Phi(\phi)$$

The eigenvalue here is $m\hbar$.

What values can m take?

For a sensible solution (standing wave)

$$\Phi(0) = \Phi(2\pi)$$

$$1 = e^{im2\pi} = \cos(m2\pi) + i \sin(m2\pi)$$

We can ignore the imaginary part as m is a real number.

$$\cos(2m\pi) = 1 \text{ implies}$$

$$m = 0, \pm 1, \pm 2, \dots$$

Polar Dependence

The polar equation is

$$-\frac{1}{\sin\theta} \frac{d}{d\theta} \left(\sin\theta \frac{d\Theta}{d\theta} \right) + \frac{m^2\Theta}{\sin^2\theta} = l(l+1)\Theta$$

The solutions to this equation are called Associated Legendre polynomials $P_l^m(\cos\theta)$.

Aside:

Equation for Associated Legendre Polynomials $P_l^m(x)$ is:

$$\frac{d}{dx} \left[(1-x^2) \frac{d}{dx} P_l^m(x) \right] + \left[l(l+1) - \frac{m^2}{1-x^2} \right] P_l^m(x) = 0$$

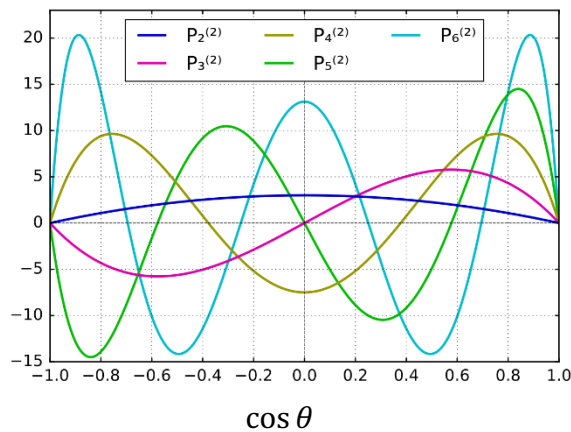
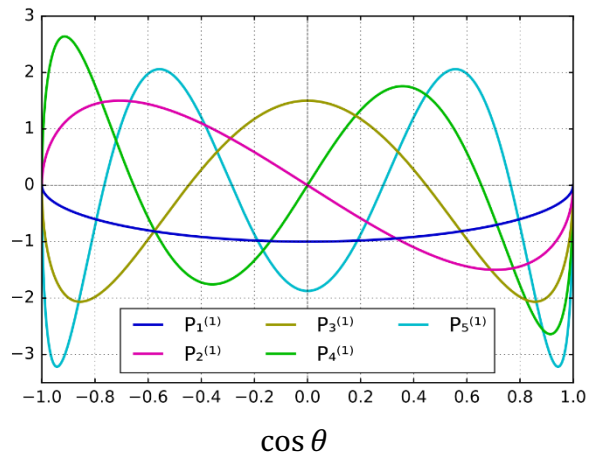
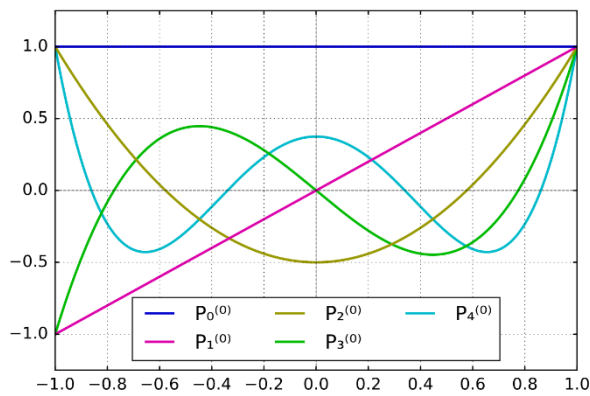
If $x = \cos \theta$ and $1-x^2 = \sin^2 \theta$

$$\frac{dx}{d\theta} = -\sin \theta$$

$$\frac{d}{dx} = \frac{d\theta}{dx} \frac{1}{d\theta} = -\frac{1}{\sin \theta} \frac{d}{d\theta}$$

$$-\frac{1}{\sin \theta} \frac{d}{d\theta} \left[-\sin \theta \frac{d}{d\theta} P_m^l(\cos \theta) \right] + \left[l(l+1) - \frac{m^2}{\sin^2 \theta} \right] P_m^l(\cos \theta) = 0$$

$$-\frac{1}{\sin \theta} \frac{d}{d\theta} \left[\sin \theta \frac{d}{d\theta} P_m^l(\cos \theta) \right] + \frac{m^2}{\sin^2 \theta} P_m^l(\cos \theta) = l(l+1) P_m^l(\cos \theta)$$



$$P_0^0(\cos \theta) = 1$$

$$P_1^0(\cos \theta) = \cos \theta$$

$$P_1^1(\cos \theta) = -(1 - \cos^2 \theta)^{\frac{1}{2}} = -\sin \theta$$

$$P_1^{-1}(\cos \theta) = -\frac{1}{2}P_1^1(\cos \theta) = \frac{1}{2}\sin \theta$$

$$P_l^{-m} = (-1)^m \frac{(l-m)!}{(l+m)!} P_m^l$$

$$P_1^{-1} = -\frac{(0)!}{(2)!} P_1^1 = \frac{1}{2}\sin \theta$$

In spherical polar coordinates, $\hat{L}^2$ depends on both θ and ϕ , whereas $\hat{L}_z$ just depends on ϕ . Due to this interdependence, the angular functions are combined. Since the two operators commute, they share a complete set of eigenfunctions.

$$Y_l^m(\theta, \phi) = N P_l^m(\cos \theta) e^{im\phi}$$

These new functions are called Spherical Harmonics.

$$N = (-1)^m \sqrt{\frac{2l+1}{4\pi} \frac{(l-m)!}{(l+m)!}}$$

$$\begin{aligned} \hat{L}^2 Y_l^m(\theta, \phi) &= l(l+1)\hbar^2 Y_l^m(\theta, \phi) \quad l = 0, 1, 2 \dots \\ \hat{L}_z Y_l^m(\theta, \phi) &= m\hbar Y_l^m(\theta, \phi) \quad m = 0, \pm 1, \pm 2, \dots \end{aligned}$$

l determines the magnitude of the orbital angular momentum of the atomic electron as it orbits the nucleus. m determines the projection of the orbital angular momentum vector along the z-axis.

The values of m and l are related (which follows from)

$$L_z^2 \leq L^2$$

$$m^2 \hbar^2 \leq l(l+1) \hbar^2$$

As both of m and l are integers then $|m| \leq l$

[check $m = l$: $l^2 \leq l(l+1)$ [ok] and $m = l+1$: $(l+1)^2 > l(l+1)$ [not ok]]

This implies that $L_z^2 < L^2$

Except when

$$L_z^2 \leq L^2$$

$$l = 0 \text{ and then } L_z^2 = L^2 = 0$$

Remember also that $\hat{L}_x$, $\hat{L}_y$ and $\hat{L}_z$ do not commute so that L cannot be aligned with the z-axis (cannot know precisely L).

Consider

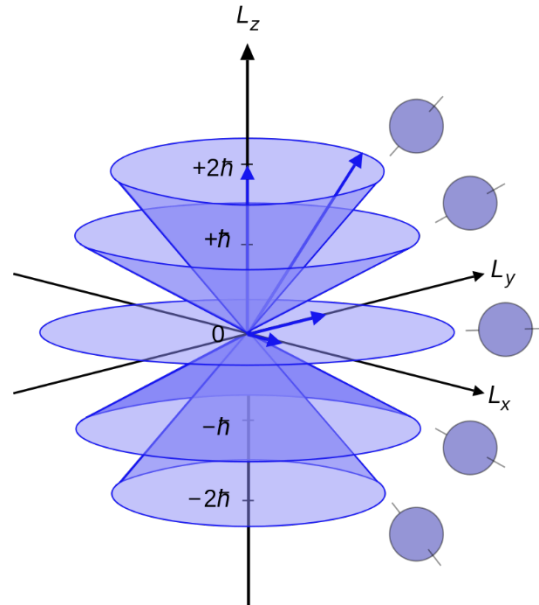
$$l = 2$$

$$L^2 = 2(2 + 1)\hbar^2$$

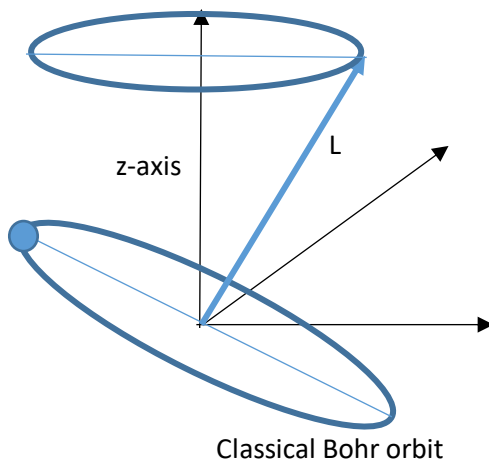
$$|L| = \sqrt{6}\hbar = 2.44\hbar$$

$$|m_l| = -2, -1, 0, 1, 2 \quad |m| \leq l$$

$$L_z = 0, \pm\hbar, \pm 2\hbar$$

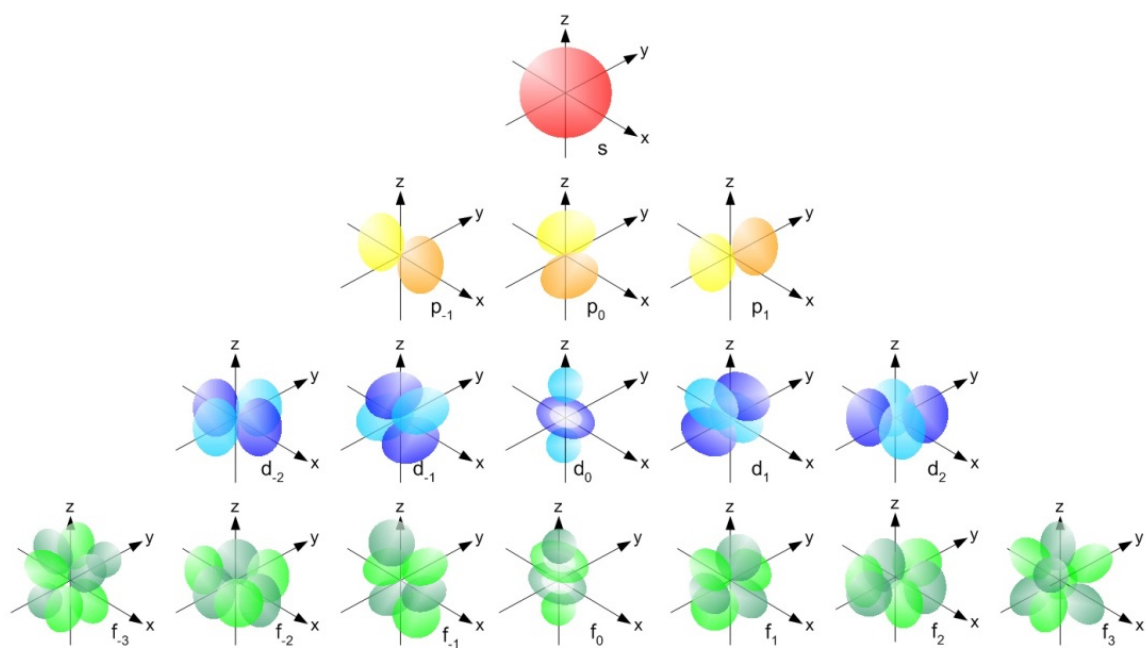
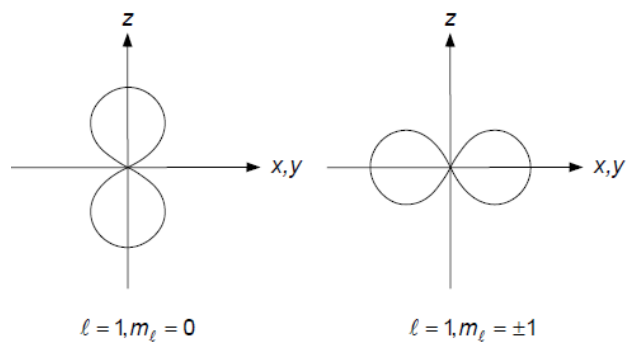


The classical view of this would be:



What do the QM orbits look like?

- i) $l = 0; m_l = 0; Y_0^0(\theta, \phi) = \left(\frac{1}{4\pi}\right)^{1/2}$
- ii) $l = 1; m_l = 0; Y_1^0(\theta, \phi) = \left(\frac{3}{4\pi}\right)^{1/2} \cos \theta$
- iii) $l = 1; m_l = 1; Y_1^1(\theta, \phi) = \left(\frac{3}{4\pi}\right)^{1/2} \sin \theta e^{i\phi}/\sqrt{2}$
- iv) $l = 1; m_l = -1; Y_1^{-1}(\theta, \phi) = -\left(\frac{3}{4\pi}\right)^{1/2} \sin \theta e^{-i\phi}/\sqrt{2}$



Radial Dependence

Only the radial equation involves energy.

$$\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \frac{2\mu}{\hbar^2} (E - V(r)) R = l(l+1) \frac{R}{r^2}$$

We find that E is quantised and there is an associated quantum number n which labels the solutions.

$$E_n = -\frac{\mu Z^2 e^4}{(4\pi\epsilon_0)^2 2\hbar^2 n^2} = -\frac{13.6}{n^2} Z^2 \text{ eV}$$

$$E_1 = -13.6 \text{ eV}$$

$$E_2 = -3.39 \text{ eV}$$

$$E_3 = -1.51 \text{ eV}$$

$$E_4 = -0.85 \text{ eV}$$

n is called the principle quantum number; is an integer.

l is the orbital angular momentum quantum number.

$$|l| = \sqrt{l(l+1)}\hbar ; l \text{ is an integer.}$$

The value of l is limited by n such that

$$0 \leq l < n$$

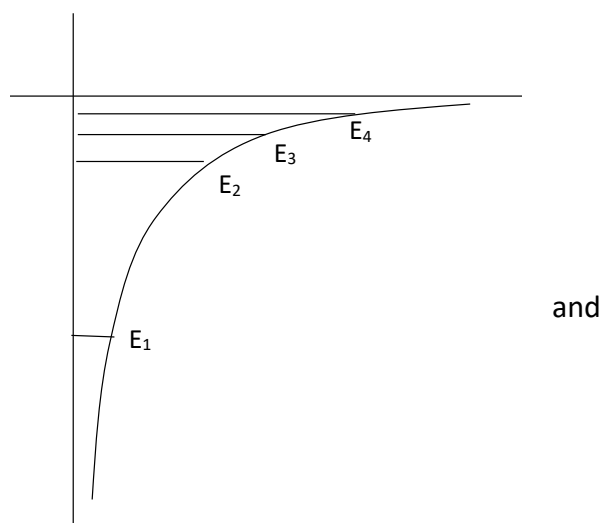
The classical interpretation of this would be the smaller n the smaller the value of $\langle r \rangle$ and the lower the value of E or p . Orbital angular momentum is $\underline{L} = \underline{r} \times \underline{p}$.

m_l is the magnetic quantum number and determines the z-component of L .

$$|m_l| \leq l ; m_l \text{ is an integer and } l_z = m_l \hbar.$$

In summary

n	$l < n$	Spectroscopic notation	$ m_l \leq l$	Degeneracy value (n^2)
1	0	1s	0	1
2	0	2s	0	4
	1	2p	$0, \pm 1$	
3	0	3s	0	9
	1	3p	$0, \pm 1$	
	2	3d	$0, \pm 1, \pm 2$	



Note the levels, e.g. 2s and 2p, are split at a fine energy scale, called fine splitting. There is something also called hyperfine splitting.

The radial solutions to the TISE have the form

$$R_{nl}(r) = e^{-\frac{Zr}{na_0}} \left(\frac{Zr}{a_0}\right)^l G_{nl}\left(\frac{Zr}{a_0}\right)$$

Where $G_{nl}\left(\frac{Zr}{a_0}\right)$ is a Laguerre polynomial function of order $n - (l + 1)$

$$a_0 = \frac{4\pi\epsilon_0\hbar^2}{\mu e^2} = 0.529 \times 10^{-10} \text{ m}$$

and is the Bohr radius.

The first few un-normalised eigenfunctions are

$$R_{10} \propto e^{-\frac{Zr}{a_0}} \quad (n = 1, l = 0)$$

$$R_{20} \propto \left(1 - \frac{Zr}{2a_0}\right) e^{-Zr/2a_0} \quad (n = 2, l = 0)$$

$$R_{21} \propto \frac{Zr}{a_0} e^{-Zr/2a_0} \quad (n = 2, l = 1)$$

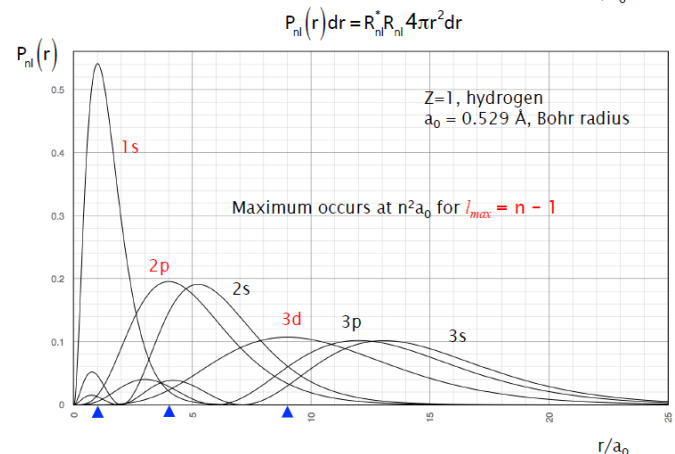
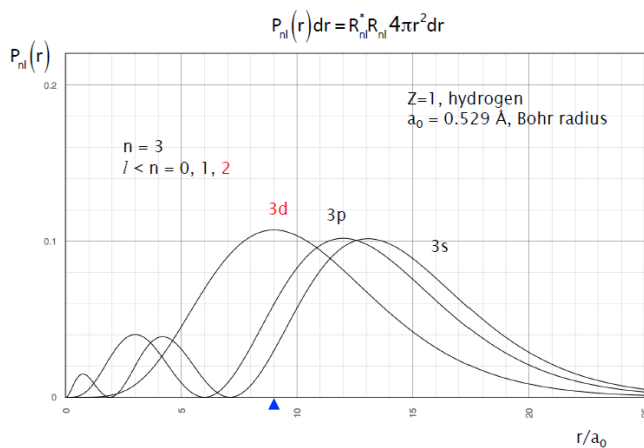
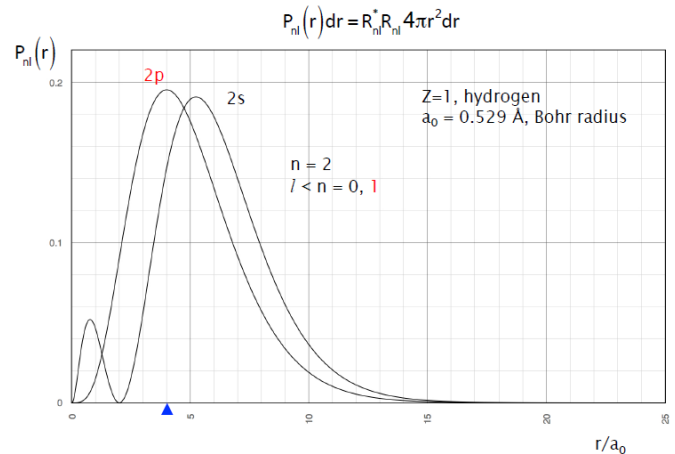
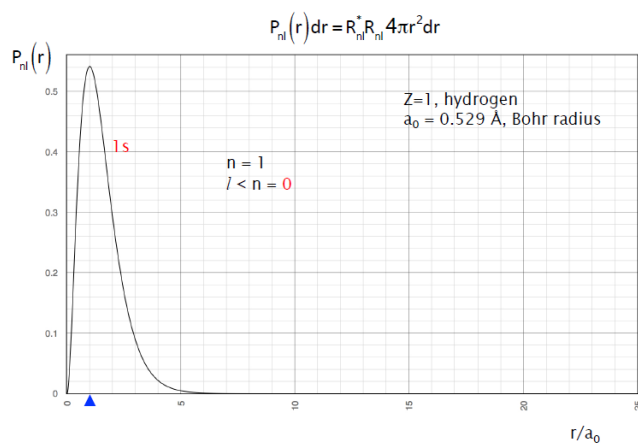
$$\frac{1}{4\pi\epsilon_0} \frac{Ze^2}{r^2} = \frac{mv^2}{r}$$

and

$$mvr = n\hbar$$

Gives

$$r = \frac{4\pi\epsilon_0\hbar^2}{mZe^2} n^2 = \frac{a_0 n^2}{Z}$$



The radial wavefunctions can be used to evaluate the radial probability density which indicates the radial distribution of the electron. The probability is

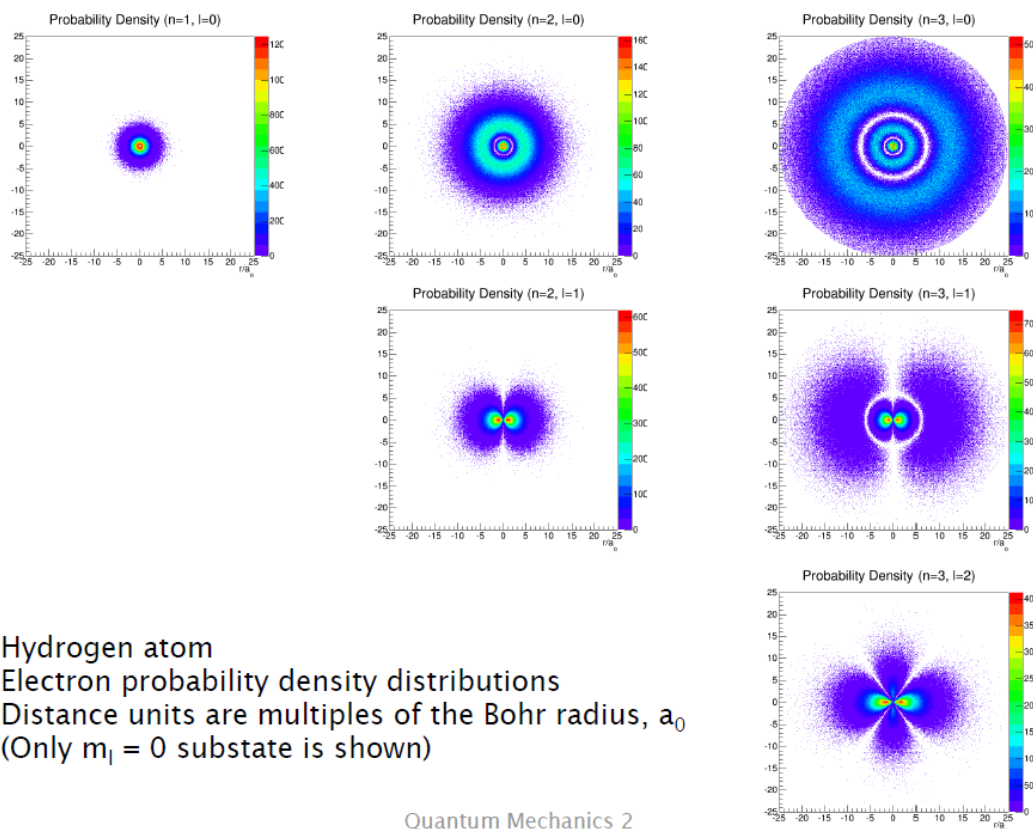
$$P(r)dr = |R_{nl}(r)|^2 \underbrace{4\pi r^2 dr}_{\text{Volume element}}$$

$$dV = dr \times r d\theta \times r \sin \theta d\phi$$

Comments:

- As n increases $\langle r \rangle$ increases
- $P(r)dr$ peaks around the Bohr radius (a_0) in the ground-state
- For $n > 1$ and $l_{max} = n - 1$ $P(r)dr$ peaks at $r = n^2 a_0$
- Exponential decay dominates as $r \rightarrow \infty$ (forbidden region)

Overall probability functions (examples):



6. Magnetic Properties and Intrinsic Spin

Many particles are associated with a property which is analogous to classical rotations or spin. For a charged object that rotates it creates a current loop and associated with that current loop is a magnetic field.

We can define a magnetic dipole moment μ .

$$\mu = I \times \text{area}$$

$$\mu = \left(-\frac{ev}{2\pi r}\right) \times \pi r^2 = -\frac{evr}{2}$$

But $\underline{l} = \underline{r} \times \underline{p}$ i.e. $|\underline{l}| = rm_e v$

$$|\underline{\mu}| = -\frac{e|\underline{l}|}{2m_e}$$

Which can be written as

$$\mu_l = -\frac{g_l \mu_B}{\hbar} l$$

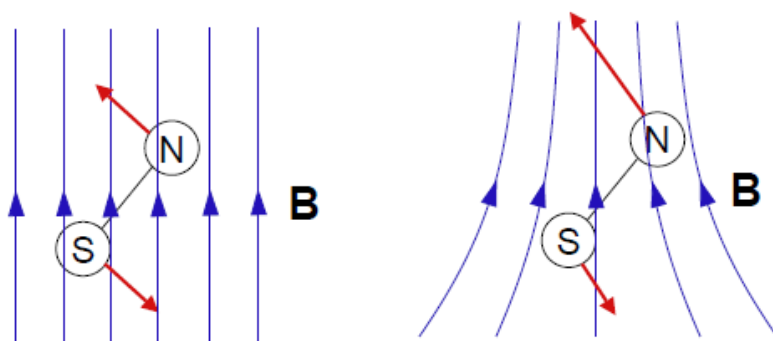
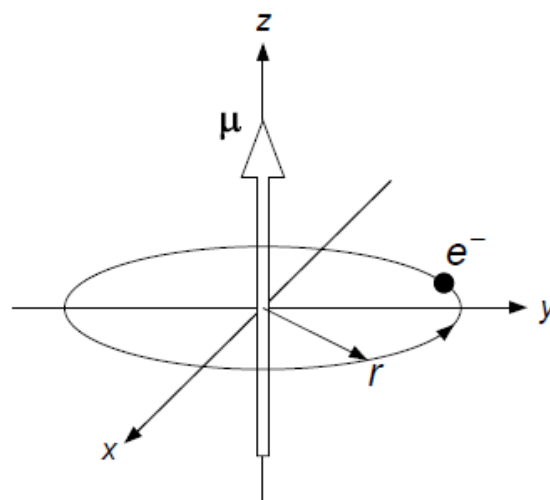
Where μ_B is called the Bohr magneton ($= \frac{e\hbar}{2m_e}$); $g_l = 1$

In QM the orbital angular momentum is quantised and just like angular momentum one can only determine $|\mu_l|$ and $(\mu_l)_z$.

$$|\mu_l| = \frac{g_l \mu_B}{\hbar} \hbar \sqrt{l(l+1)}$$

$$(\mu_l)_z = \frac{g_l \mu_B}{\hbar} \hbar m_l = g_l \mu_B m_l$$

To measure the value of $(\mu_l)_z$ one needs a non-uniform magnetic field



Here the force is related to the magnetic field gradient.

$$F \propto m_e \frac{\partial B_z}{\partial z}$$

The potential energy of a dipole is

$$V = -\boldsymbol{\mu} \cdot \mathbf{B}$$

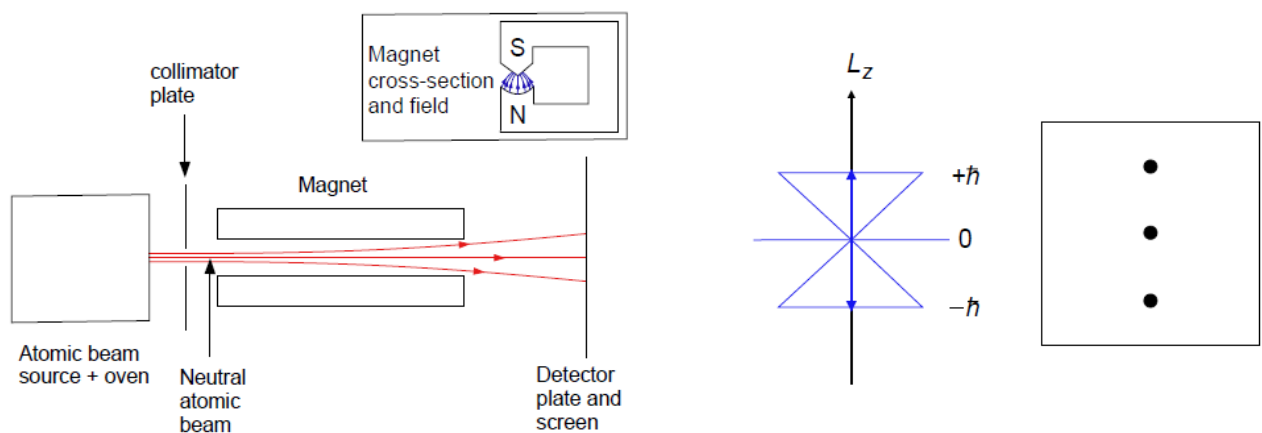
So the potential energy of the dipole in the z-direction is

$$V = g_l \mu_B m_l B_z \quad [g_l = 1]$$

Force

$$F_z = -\frac{\partial V}{\partial z} = -g_l \mu_B m_l \frac{\partial B_z}{\partial z}$$

Stern-Gerlach Experiment (for a S=1 case)



Consider a hydrogen atom in the ground state. Here

$$n = 1, l = 0, m_l = 0$$

So there is no magnetic moment from the orbital motion of the electron and hence no magnetic moment is expected and no deflection of the beam.

However, the H atom beam is split into two beams, with two spots, influenced by the spin of the electron.

The spin of the charged particle also creates an intrinsic magnetic moment (which for the hydrogen atom is dominated by the electron rather than the proton whose effect is very small).

Angular Momentum Rules for Spin

$$|S| = \sqrt{s(s+1)}\hbar$$

$$S_z = m_s \hbar$$

For the electron $s = 1/2$ and $m_s = \pm 1/2$

So there are two components of electron spin; up or down wrt the magnetic field gradient direction.

$$(\mu_s)_z = -g_s \mu_B m_s$$

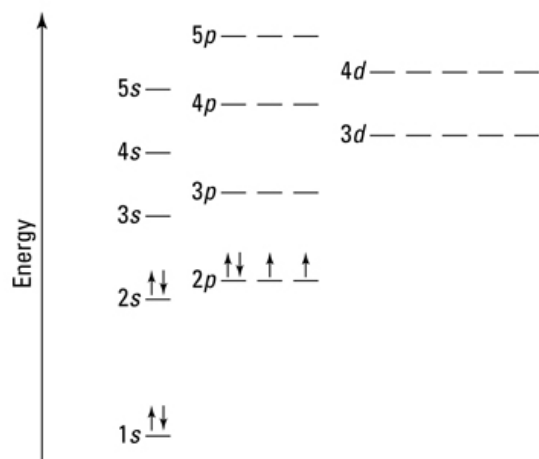
Electrons are fermions – they have half integer spin and obey the Pauli Exclusion Principle. This means that no two fermions can occupy the same quantum state.

Since for each orbital quantum state there are two possible spin orientations, then each state has a degeneracy of 2 ($g_l = 2$).

Atom	n	l	m_l	m_s	# electrons
H-He	1	0	0	$\pm 1/2$	2
Li-Ne	2	0	0	$\pm 1/2$	2
		1	$0, \pm 1$	$\pm 1/2$	6
Na- ... etc.	3	0	0	$\pm 1/2$	2
		1	$0, \pm 1$	$\pm 1/2$	6
		2	$0, \pm 1, \pm 2$	$\pm 1/2$	10

Number of electrons is given by $2(2l + 1)$. We can build the Periodic Table !

Due to relativistic effects and spin-orbit interaction (see QM3), levels with different orbital ang. mom. have different energies. Considering this, see example with 8 electrons:



We can combine the orbital angular momentum and spin into a total angular momentum $\underline{J}$

$$\underline{J} = \underline{L} + \underline{S}$$

$\underline{J}$ is a vector and can take values $|l - s| \leq j \leq l + s$

For an electron, $s=1/2$ and hence $j = l + 1/2$ or $j = l - 1/2$.

$$|\underline{J}| = \sqrt{j(j+1)}\hbar$$

$$J_z = m_j \hbar$$

There are thus two ways to refer to the quantum states

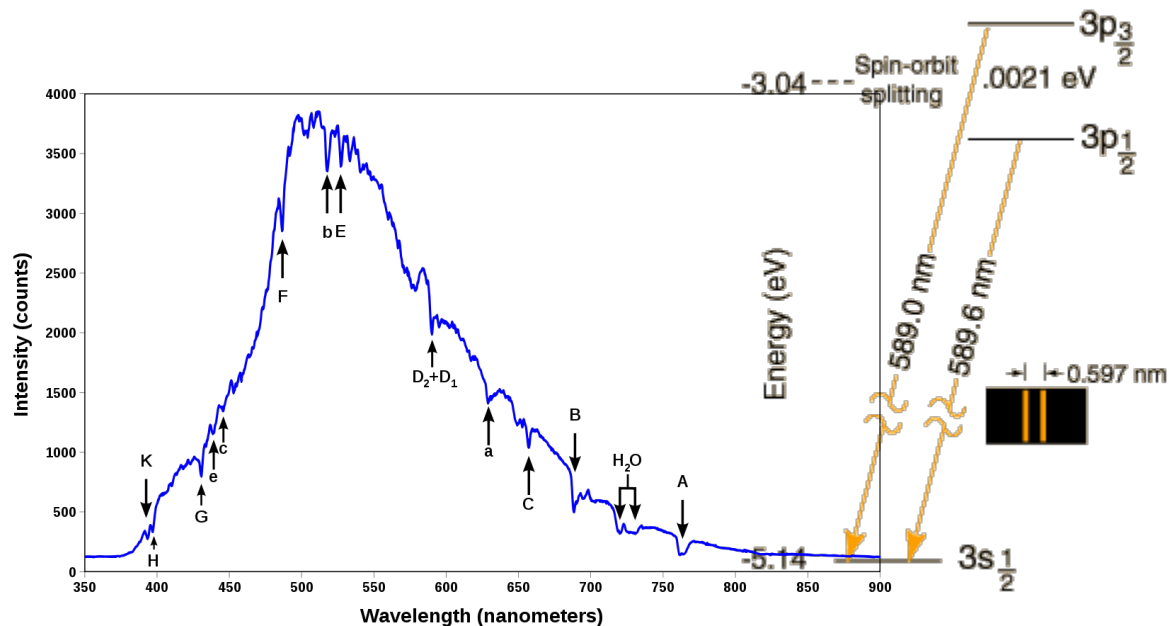
Either use: n, l, m_l and m_s

Or: n, l, j and m_j

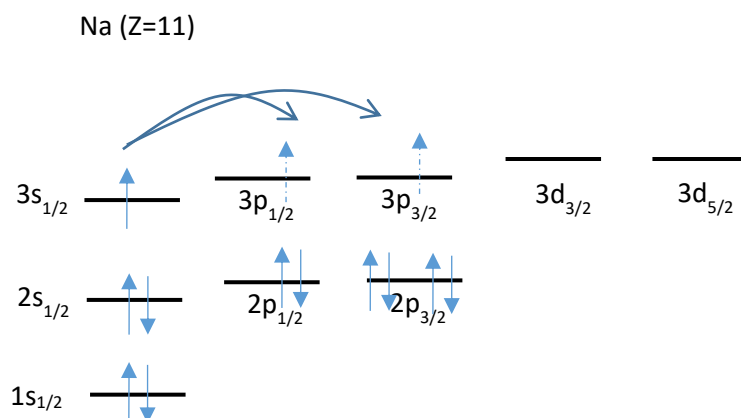
So we can label the orbits now with quantum numbers j . This is the preferred notation.

$l = 0$	$j = 1/2$	$m_j = \pm 1/2$	$s_{1/2}$
$l = 1$	$j = 3/2$	$m_j = \pm 1/2, \pm 3/2$	$p_{3/2}$
	$j = 1/2$	$m_j = \pm 1/2$	$p_{1/2}$
$l = 2$	$j = 5/2$	$m_j = \pm 1/2, \pm 3/2, \pm 5/2$	$d_{5/2}$
	$j = 3/2$	$m_j = \pm 1/2, \pm 3/2$	$d_{3/2}$

$2j + 1$ magnetic substates.



Splitting of the atomic levels into j -quantum numbers is fine structure, you will see this in future courses. Seen for example in Sodium “D-lines”. First observed in solar spectrum.



7. Ladder Operators

Angular momentum

Spherical harmonics for a given angular momentum can be in the range Y_l^l to Y_l^{-l} . Ladder operators can be used to move between these eigenstates. $\hat{L}_+$ increases the value of m , and $\hat{L}_-$ decreases the value of m .

$$\hat{L}_+ = \hat{L}_x + i\hat{L}_y$$

$$\hat{L}_- = \hat{L}_x - i\hat{L}_y$$

Note that neither of these operators are Hermitian, so they don't have real eigenvalues.

Commutators:

$$[\hat{L}_+, \hat{L}_-] = [\hat{L}_x + i\hat{L}_y, \hat{L}_x - i\hat{L}_y] = -i \underbrace{[\hat{L}_x, \hat{L}_y]}_{i\hbar\hat{L}_z} + i \underbrace{[\hat{L}_y, \hat{L}_x]}_{-i\hbar\hat{L}_z} = 2\hbar\hat{L}_z \quad (\text{Do not commute})$$

$$[\hat{L}_z, \hat{L}_\pm] = [\hat{L}_z, \hat{L}_x \pm i\hat{L}_y] = \underbrace{[\hat{L}_z, \hat{L}_x]}_{i\hbar\hat{L}_y} \pm i \underbrace{[\hat{L}_z, \hat{L}_y]}_{-i\hbar\hat{L}_x} = i\hbar\hat{L}_y \pm \hbar\hat{L}_x = \pm\hbar\hat{L}_\pm \quad (\text{Do not commute})$$

But since $[\hat{L}^2, \hat{L}_x] = 0$ and $[\hat{L}^2, \hat{L}_y] = 0$, then $[\hat{L}^2, \hat{L}_\pm] = 0$, i.e. commutes.

This means applying the ladder operators to a state does not change the eigenvalue of $\hat{L}^2$.

But does change $\hat{L}_z$.

$$\text{Starting from } [\hat{L}_z, \hat{L}_\pm] = \hat{L}_z\hat{L}_\pm - \hat{L}_\pm\hat{L}_z = \pm\hbar\hat{L}_\pm \quad (\text{above})$$

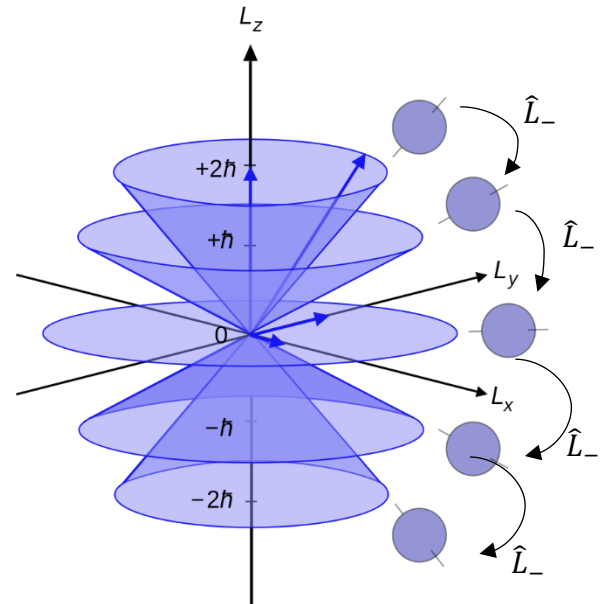
$$\hat{L}_z\hat{L}_\pm = \hat{L}_\pm\hat{L}_z \pm \hbar\hat{L}_\pm$$

$$\hat{L}_z\hat{L}_\pm Y_l^m = (\hat{L}_\pm\hat{L}_z \pm \hbar\hat{L}_\pm) Y_l^m = \hbar(m \pm 1)\hat{L}_\pm Y_l^m$$

So $\hat{L}_z$ operating on the wavefunction after $\hat{L}_\pm$ returns not $m\hbar$ but $(m \pm 1)\hbar$.

$$\hat{L}_\pm Y_l^m = c_{lm}^\pm Y_l^{m\pm 1} \quad (c_{lm}^\pm \text{ are complex numbers})$$

Note that $\hat{L}_\pm$ cannot raise, or lower, the value of m above l or below $-l$.



Harmonic Oscillator

Remember for HO

$$\hat{H}\psi = \frac{1}{2m}(\hat{p}^2 + m^2\omega^2 x^2)\psi = E\psi ; \hat{p} = -i\hbar \frac{\partial}{\partial x}$$

Consider two operators

$$\hat{a}_+ = \frac{1}{\sqrt{2\hbar m\omega}}(-i\hat{p} + m\omega\hat{x})$$

and

$$\hat{a}_- = \frac{1}{\sqrt{2\hbar m\omega}}(i\hat{p} + m\omega\hat{x})$$

Then

$$\begin{aligned}\hat{a}_+\hat{a}_- &= \frac{1}{\sqrt{2\hbar m\omega}}(-i\hat{p} + m\omega\hat{x}) \frac{1}{\sqrt{2\hbar m\omega}}(i\hat{p} + m\omega\hat{x}) \\ &= \frac{1}{2\hbar m\omega}(\hat{p}^2 + m^2\omega^2\hat{x}^2 + im\omega(\hat{x}\hat{p} - \hat{p}\hat{x}))\end{aligned}$$

$$\begin{aligned}\hat{a}_-\hat{a}_+ &= \frac{1}{\sqrt{2\hbar m\omega}}(i\hat{p} + m\omega\hat{x}) \frac{1}{\sqrt{2\hbar m\omega}}(-i\hat{p} + m\omega\hat{x}) \\ &= \frac{1}{2\hbar m\omega}(\hat{p}^2 + m^2\omega^2\hat{x}^2 - im\omega(\hat{x}\hat{p} - \hat{p}\hat{x}))\end{aligned}$$

Noting that $[\hat{x}, \hat{p}] = i\hbar$

$$\hat{a}_+\hat{a}_- = \frac{1}{\hbar\omega} \left[\hat{H} - \frac{1}{2}\hbar\omega \right] = \frac{\hat{H}}{\hbar\omega} - \frac{1}{2}$$

$$\hat{a}_-\hat{a}_+ = \frac{1}{\hbar\omega} \left[\hat{H} + \frac{1}{2}\hbar\omega \right] = \frac{\hat{H}}{\hbar\omega} + \frac{1}{2}$$

So the TISE can be rewritten as

$$\left(\hat{a}_+\hat{a}_- + \frac{1}{2} \right) \psi = \frac{E}{\hbar\omega} \psi$$

or

$$\left(\hat{a}_-\hat{a}_+ - \frac{1}{2} \right) \psi = \frac{E}{\hbar\omega} \psi$$

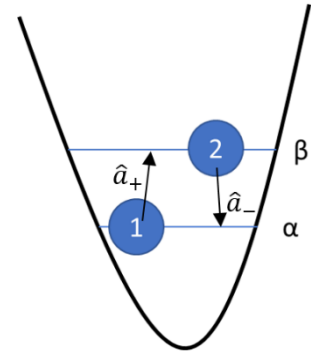
Now consider

$$\hat{H}(\hat{a}_+\psi) = \hbar\omega \left(\hat{a}_+\hat{a}_- + \frac{1}{2} \right) \hat{a}_+\psi = \hbar\omega \left(\hat{a}_+\hat{a}_-\hat{a}_+\psi + \frac{1}{2}\hat{a}_+\psi \right) = \hbar\omega \hat{a}_+ \left(\hat{a}_-\hat{a}_+\psi + \frac{1}{2}\psi \right)$$

which gives

$$\begin{aligned}\hat{H}(\hat{a}_+\psi) &= \hbar\omega\hat{a}_+\left(\frac{\hat{H}}{\hbar\omega}\psi + \frac{1}{2}\psi + \frac{1}{2}\psi\right) = \hat{a}_+(\hat{H}\psi + \hbar\omega\psi) \\ &= \hat{a}_+(E\psi + \hbar\omega\psi) = (E + \hbar\omega)\hat{a}_+\psi\end{aligned}$$

Thus similar to angular momentum ladder operators..... this implies that if one operates with the $\hat{a}_+$ operator on the wavefunction and then measures the energy of the state it is $1\hbar\omega$ higher than before. Similarly, the $\hat{a}_-$ operator would give $1\hbar\omega$ lower than before.



Since

$$\hat{a}_-\psi_0 = 0 \text{ (cannot go below lowest state)}$$

$$\hat{H}\psi_0 = \hbar\omega\left(\hat{a}_+\hat{a}_- + \frac{1}{2}\right)\psi_0 = \hbar\omega\left(\hat{a}_+\hat{a}_-\psi_0 + \frac{1}{2}\psi_0\right) = \frac{1}{2}\hbar\omega\psi_0$$

So

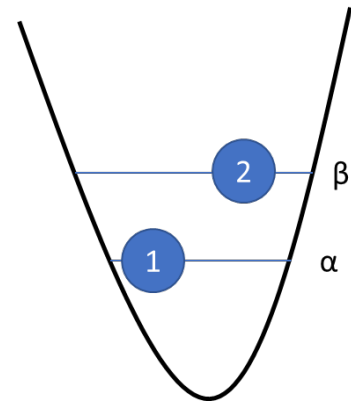
$$E_0 = \frac{1}{2}\hbar\omega$$

And thus

$$E_n = \frac{1}{2}\hbar\omega + n\hbar\omega = \left(n + \frac{1}{2}\right)\hbar\omega$$

8. Multiple Particles: preview of topics in future QM courses

Consider two identical non-interacting particles, particle 1 and particle 2, in a potential with two levels α and β .



$$\hat{H}_1\psi_1(\alpha) = E_1\psi_1(\alpha)$$

$$\hat{H}_2\psi_2(\beta) = E_2\psi_2(\beta)$$

Energies should sum i.e.

$$\hat{H}_{12} = \hat{H}_1 + \hat{H}_2 \text{ and}$$

$$E_{12} = E_1 + E_2$$

$$\hat{H}_{12}\psi_{12} = (E_1 + E_2)\psi_{12}$$

But what is the right form of the two-particle wavefunction?

$$\text{Try } \psi_{12} = \psi_1 + \psi_2$$

$$\hat{H}_{12}\psi_{12} = (\hat{H}_1 + \hat{H}_2)(\psi_1 + \psi_2) = \hat{H}_1\psi_1 + \underbrace{\hat{H}_2\psi_1}_{=0} + \underbrace{\hat{H}_1\psi_2}_{=0} + \hat{H}_2\psi_2 = E_1\psi_1 + E_2\psi_2 \neq E\psi_{12}$$

What about $\psi_{12} = \psi_1\psi_2$?

$$\hat{H}_{12}\psi_{12} = \hat{H}_1\psi_1\psi_2 + \hat{H}_2\psi_1\psi_2 = E_1\psi_1\psi_2 + E_2\psi_1\psi_2 = (E_1 + E_2)\psi_1\psi_2 \text{ OK!}$$

How should we write $\psi_1\psi_2$ as the particles are identical?

$$\psi_1\psi_2 = \psi_1(\alpha)\psi_2(\beta) \text{ or } = \psi_2(\alpha)\psi_1(\beta)$$

Both are the same. Result will then be a linear combination of both

$$\psi_1\psi_2 = \frac{1}{\sqrt{2}}(\psi_1(\alpha)\psi_2(\beta) \pm \psi_2(\alpha)\psi_1(\beta))$$

Let's introduce a permutation operator to swap particles

$$\hat{P}_{12}\psi_{12} = \lambda\psi_{12}$$

$$\hat{P}_{12}(\hat{P}_{12}\psi_{12}) = \lambda^2\psi_{12} = \psi_{12}$$

$$\lambda^2 = 1 \text{ or } \lambda = \pm 1$$

So either there is a no change of sign (symmetric) or a change of sign (antisymmetric) when the particles are swapped

Symmetric (bosons):

$$\psi_1\psi_2 = \frac{1}{\sqrt{2}}(\psi_1(\alpha)\psi_2(\beta) + \psi_2(\alpha)\psi_1(\beta))$$

Antisymmetric (fermions):

$$\psi_1\psi_2 = \frac{1}{\sqrt{2}}(\psi_1(\alpha)\psi_2(\beta) - \psi_2(\alpha)\psi_1(\beta))$$

For fermions if two particles are in the same state

$$\psi_1\psi_2 = \frac{1}{\sqrt{2}}(\psi_1(\alpha)\psi_2(\alpha) - \psi_2(\alpha)\psi_1(\alpha)) = 0$$

The wavefunction vanishes – this is the Pauli Exclusion Principle.